

#1

$$* \text{ added table salt} = \frac{3}{4} \text{ tsp} \times 6 \text{ g/tsp} = 4.50 \text{ g/NaCl}$$

$$\begin{aligned} * \text{ Sodium from that salt: } & 4.50 \text{ g} \times \frac{M_{\text{Na}}}{M_{\text{NaCl}}} \\ & = 4.50 \times \frac{22.9898}{58.442} \\ & = 1770 \text{ mg Na} \end{aligned}$$

* baseline sodium in 1lb (16 oz) of unsalted peanuts

$$= 5 \text{ mg} \times 16 = 80 \text{ mg Na}$$

$$\text{total Na} = 1770 + 80 = 1850 \text{ mg Na}$$

a) lightly salted peanuts

$$* 75 \times 16 = 1200 \text{ mg Na}$$

$$* 1850 - 1200 = 650 \text{ mg Na}$$

$$\frac{650}{22.9898 / 58.442} = \frac{1.432 \text{ g}}{6.0} = 0.28 \text{ tsp}$$

b) Salted peanuts

$150 \times 16 = 2400 \text{ mg Na}$
which already exceeds the 1850.

add 0 tsp



c) lightly salted peanuts can be adjusted by adding 0.28 tsp of salt. but you can't add anything to the salted peanuts b/c it's already too salty

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#2 / total product = 200 g ✓

$$n_{\text{Cu}} = \frac{100}{63.546} = 1.57366 \text{ mol}$$

$$n_{\text{Cl}} = \frac{100}{35.543} = 2.82064 \text{ mol}$$

$$* 2.82064 - 1.57366 = 1.24698 \text{ mol}$$

$$* 1.57366 - 1.24698 = 0.32669 \text{ mol}$$

Convert to masses

$$M_{\text{CuCl}_2} = 134.452, \quad M_{\text{CuCl}} = 98.99 \text{ g/mol}$$

$$m(\text{CuCl}_2) = 1.24698 \times 134.452 = 167.658 \text{ g}$$

$$m(\text{CuCl}) = 0.32669 \times 98.999 = 32.342 \text{ g}$$

$$167.658 + 32.342 = 200 \text{ g} \quad \checkmark$$

$$\text{Mass ratio} = 5.18 : 1 \quad \checkmark$$

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#3] $C_p(T) = A + BT + CT^2 + DT^3$ ✓

$A = 28.11$, $B = -3.68 \times 10^{-6}$, $C = 1.746 \times 10^{-5}$, $D = -1.065 \times 10^{-8}$

a) $\Delta h = A(T_2 - T_1) + \frac{B}{2}(T_2^2 - T_1^2) + \frac{C}{3}(T_2^3 - T_1^3) + \frac{D}{4}(T_2^4 - T_1^4)$
 $= 28.11(700 - 300) + \frac{-3.68 \times 10^{-6}}{2}(700^2 - 300^2) + \frac{1.746 \times 10^{-5}}{3}(700^3 - 300^3) + \frac{-1.065 \times 10^{-8}}{4}(700^4 - 300^4)$ ✓

$\Delta h = 1.246 \times 10^4 \text{ J/mol}$ ✓

b) $C_p(T) = A + (1000B)t + (10^6C)t^2 + (10^9D)t^3$ ✓
 $= 28.11 + (1000 \times -3.68 \times 10^{-6})t + (10^6 \times 1.746 \times 10^{-5})t^2 + (10^9 \times -1.065 \times 10^{-8})t^3$
 $= 1.246 \times 10^4 \text{ J/mol}$ ✓

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#4) $I(h) = \int_1^h \frac{1}{40-4\sqrt{x-1}} dx, \quad (1 \leq h \leq 10)$

* let $u = \sqrt{x-1}$, $x = u^2 + 1$, $dx = 2u du$
 $x = 1 \rightarrow u = 0$, $x = h \rightarrow u = \sqrt{h-1}$

$$I(h) = \int_0^{\sqrt{h-1}} \frac{2u}{40-4u} du = \int_0^{\sqrt{h-1}} \frac{1}{2} \frac{u}{10-u} du = \frac{1}{2} \int_0^{\sqrt{h-1}} \frac{u}{10-u} du$$

$$\frac{u}{10-u} = \frac{-(10-u) + 10}{10-u} = -1 + \frac{10}{10-u}$$

$$\Rightarrow I(h) = \frac{1}{2} \int_0^{\sqrt{h-1}} \left(-1 + \frac{10}{10-u} \right) du = \frac{1}{2} \left[u - 10 \ln |10-u| \right]_0^{\sqrt{h-1}}$$

$$\Rightarrow I(h) = \frac{1}{2} \left[-\sqrt{h-1} - 10 \ln(10 - \sqrt{h-1}) + 10 \ln(10) \right]$$

$$\int_1^h \frac{dx}{40-4\sqrt{x-1}} = -\frac{\sqrt{h-1}}{2} - 5 \ln(10 - \sqrt{h-1}) + 5 \ln(10)$$

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#5) Alcohol = 7 kcal g⁻¹, molar mass (ethanol): $M = 46.07 \text{ g mol}^{-1}$

Convert to molar energy:

$$7 \times 46.07 = 322.49 \text{ kcal/mol}$$

$$322.49 \times 4.184 = 1349.3$$

* liquid ethanol = -1367 $\rightarrow$ magnitude differs by $1367 - 1349.3$
= 17.7 kJ/mol

* gas ethanol: -1235 $\rightarrow$ 9.3% lower in magnitude than 1349.3

the dietary 7 kcal g⁻¹ matches the magnitude of the standard enthalpy of combustion of liquid ethanol very closely. nutrition labels report energy to you as a positive number, while thermodynamics reports the reaction heat as negative (exothermic)

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